

Final Report – Credit Risk Classification

Using Machine Learning

The purpose of this project was to use machine learning techniques to predict credit risk using a real-world financial dataset. Credit risk is an important problem for banks and lenders because they need to determine whether a borrower is likely to repay a loan or default on it. By using machine learning models, financial institutions can make faster and more accurate lending decisions while reducing risk. The main goal of this project was to compare different classification models and determine which one performed best when predicting loan risk.

The dataset used for this project was the Credit Risk Dataset from Kaggle. The dataset included multiple financial and demographic variables such as age, income, employment length, loan amount, loan interest rate, home ownership status, and loan intent. The target variable for the project was loan status, which represents whether a borrower defaulted on a loan or not. Since the target variable only had two outcomes, this project was treated as a binary classification problem.

Before training the models, the dataset went through several preprocessing steps that were originally completed during the midterm project. Missing values were handled using imputation techniques, and unrealistic outliers were identified and corrected. Categorical variables were converted into numerical form using encoding methods so that machine learning algorithms could process the data correctly. Feature scaling and normalization were also applied using MinMaxScaler to keep the numerical values within a similar range. In addition, a debt-to-

income ratio feature was created to provide a better representation of a borrower's financial situation.

After preprocessing, the dataset was split into training, validation, and testing datasets using a 70%, 15%, and 15% split. The training data was used to train the models, while the validation and test sets were used to evaluate performance on unseen data. Several machine learning classification models were trained during the project, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, K-Nearest Neighbors (KNN), Support Vector Classifier (SVC), and Gaussian Naive Bayes.

Each model was evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. These metrics helped compare how well the models handled classification predictions. Logistic Regression produced decent results but had lower recall and F1-scores compared to the stronger ensemble-based models. Since Logistic Regression assumes linear relationships between variables, it struggled more with the complex patterns found in financial data.

The Decision Tree model performed better than Logistic Regression and was able to capture more nonlinear relationships in the dataset. However, Decision Trees can sometimes overfit training data, which can reduce performance consistency on new data. KNN and SVC also produced respectable results, but their recall and F1-scores were lower than the top-performing models.

The strongest performing model overall was the Random Forest Classifier. Random Forest achieved the highest F1-score and one of the best ROC-AUC scores on both the validation and test datasets. This model performed especially well because it combines multiple decision

trees together, reducing overfitting while improving overall prediction accuracy. Random Forest was able to better capture the complex relationships between financial variables, making it more effective at identifying risky borrowers.

Gradient Boosting also performed very well and had results close to Random Forest. Its sequential boosting process helped improve weak predictions over time, leading to strong accuracy and ROC-AUC scores. Although it slightly underperformed Random Forest in F1-score, it still demonstrated excellent classification performance overall.

In addition to the individual models, an ensemble Voting Classifier was created using some of the top-performing models. The Voting Ensemble combined Random Forest, Gradient Boosting, and SVC together using soft voting. The ensemble model achieved strong overall performance and improved prediction stability by combining the strengths of multiple classifiers. Ensemble learning helped reduce weaknesses from individual models and produced highly reliable results on the test dataset.

A Bayesian classification model using Gaussian Naive Bayes was also included for comparison. While the Bayesian model performed reasonably well, its overall metrics were lower than the ensemble-based methods. One reason for this is that Gaussian Naive Bayes assumes independence between features, while many financial variables in the dataset are naturally related to one another.

Overall, this project demonstrated how machine learning can be used to solve real-world financial classification problems. The results showed that ensemble-based methods such as Random Forest and Voting Classifiers performed best for predicting credit risk. These models

were able to provide strong accuracy, balanced precision and recall, and high ROC-AUC scores.

The project also showed the importance of proper preprocessing, feature engineering, and model comparison when building machine learning systems.